

Network Traffic Analysis : Wireshark vs. tcpdump

1. Project overview

In this activity, I compared two widely used network protocol analyzers: Wireshark and tcpdump. The objective was to research their capabilities, identify key similarities and differences, and evaluate how each tool can be used to capture and analyze network traffic for security monitoring and troubleshooting.

2. Key activities

- researched the functionality of Wireshark and tcpdump
- Compared graphical and command-line packet analysis tools
- Identified common use cases for network traffic inspection
- Evaluated resource requirements and operational differences
- Created a comparison diagram highlighting strengths and limitations of each tool.

3. Key Findings

- both Wireshark and tcpdump are open-source network protocol analyzers used to capture and inspect network traffic
- Both tools support protocol filtering and packet analysis for troubleshooting and security investigations.
- Wireshark provides a graphical interface with advanced filtering, protocol decoding, and visualization features.
- Tcpdump operates through the command-line and lightweight, making it suitable for remote systems and servers with limited resources.

4. Lesson learned

- different cybersecurity tools may achieve similar objectives while serving different operational needs
- GUI-based tools like Wireshark simplify packet analysis and are useful during investigations and learning activities
- Command-line tools such as tcpdump are often preferred in Linux environments and production systems because of their efficiency and low resource consumption
- Understanding both tools improves an analyst's ability to investigate network-related security incident across different environment.

5. Skill demonstrated

- network traffic analysis
- Packet inspection
- Wireshark
- Tcpdump
- Network protocol analysis
- Security monitoring
- Research and documentation
- Technical comparison and evaluation